

Multiple Choice

1. **Answer:** D

Options: A) Momentum is known exactly, but no information about position can be known.; B) Position is known exactly, but no information about momentum can be known.; C) No information about either position or momentum can be known.; D) Neither position nor momentum can be known exactly.

Note: Correct option: D - Neither position nor momentum can be known exactly.

2. **Answer:** D

Options: A) Both ^{35}Cl and ^{37}Cl have $I = 0$.; B) The hydrogen atom undergoes rapid intermolecular exchange.; C) The molecule is not rigid.; D) Both ^{35}Cl and ^{37}Cl have electric quadrupole moments.

Note: Correct option: D - Both ^{35}Cl and ^{37}Cl have electric quadrupole moments.

3. **Answer:** C

Options: A) Protons and neutrons have orbital and spin angular momentum.; B) Protons have orbital and spin angular momentum, neutrons have spin angular momentum.; C) Protons and neutrons possess orbital angular momentum only.; D) Protons and neutrons possess spin angular momentum only.

Note: Correct option: C - Protons and neutrons possess orbital angular momentum only.

4. **Answer:** B

Options: A) HCl / NaOH ; B) $\text{H}_3\text{O}^+ / \text{H}_2\text{O}$; C) $\text{O}_2 / \text{H}_2\text{O}$; D) H^+ / Cl^-

Note: Correct option: B - $\text{H}_3\text{O}^+ / \text{H}_2\text{O}$

5. **Answer:** D

Options: A) The most common oxidation state for the lanthanide elements is +3.; B) Lanthanide complexes often have high coordination numbers (> 6).; C) All of the lanthanide elements react with aqueous acid to liberate hydrogen.; D) The atomic radii of the lanthanide elements increase across the period from La to Lu.

Note: Correct option: D - The atomic radii of the lanthanide elements increase across the period from La to Lu.

True / False

6. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

7. **Answer:** False
Options: A) True; B) False
Note: LLM-generated false statement. Correct answer: '2.9 T'.
8. **Answer:** True
Options: A) True; B) False
Note: LLM-generated true statement based on MCQ.
9. **Answer:** True
Options: A) True; B) False
Note: LLM-generated true statement based on MCQ.
10. **Answer:** False
Options: A) True; B) False
Note: LLM-generated false statement. Correct answer: 'NH 2-'.

Long Form Response

11. **Answer:** 1. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.
Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.
12. **Answer:** First ionization energy. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.
Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

End of Answer Key